NASA SPACE APPS CHALLENGE 2026

ORION SPACE

NEIA - NASA EARTH INTELLIGENT AGENT

Turning Earth-observation data into understandable, traceable and locally relevant insights.

PROBLEM STATEMENT

Be an Earth System Trend Detective

TEAM ORION SPACE • BANGLADESH

“AI explains the result; the scientific pipeline produces it.”

01 / WHY THIS MATTERS

The gap between Earth data and local action

Satellite observations are powerful,but turning them into a decision-ready explanation takes specialist tools,domain knowledge and careful interpretation.

THE SIGNAL

Rainfall, land temperature, vegetation and soil moisture reveal changes in the Earth system.

THE COMPLEXITY

Data formats, projections, scale differences, missing observations and statistical methods create friction.

THE LAST MILE

Local communities and decision-makers need clear context, not just raw satellite pixels.

THE ORION OPPORTUNITY

Make it possible to ask a question in everyday language, run a transparent geospatial analysis, and receive an answer that states the evidence, assumptions, limitations and next steps.

02 / WHAT WE ARE BUILDING

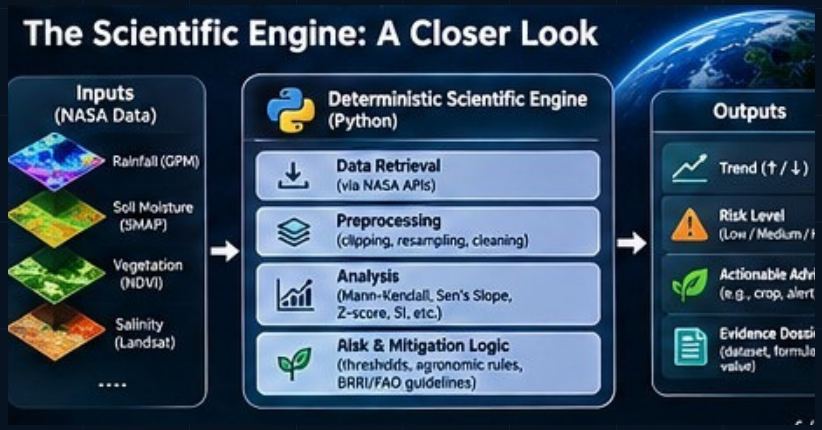
Orion Space: a glass-box AI agent

- **Conversational access**
  Ask about a place, time period, variable or observed trend.

- **Scientific computation**
  Python retrieves, cleans, aggregates and analyzes the selected data.

- **Evidence with every answer**
  Show source, time window, units, method, result and uncertainty.

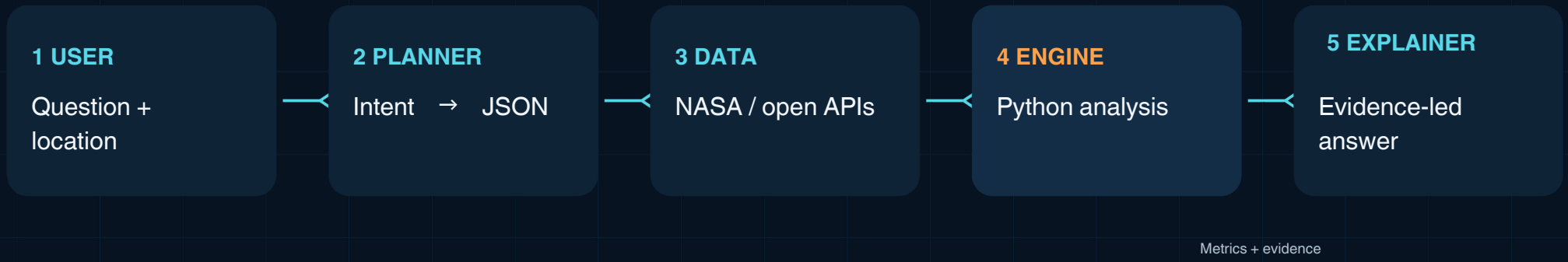
- **Decision support**
  Translate a verified signal into practical questions and locally validated options.



Illustrative interface concept

03 / END-TO-END ARCHITECTURE

From a natural-language question to a verifiable result



CROSS-CUTTING CONTROLS

Input validation • dataset provenance • unit consistency • missing-data checks • spatial boundary checks • reproducible parameters • uncertainty reporting • safe wording

04 / EARTH-OBSERVATION INPUTS

Choose datasets to match the question

GPM IMERG	Precipitation estimates	Rainfall totals, seasonal comparisons, anomalies
MODIS	Land-surface temperature, vegetation indices	Thermal stress, vegetation condition and seasonal change
SMAP	Soil moisture products	Surface / root-zone moisture context and drought monitoring
Landsat 8/9	Multispectral surface reflectance	Land-cover and surface-condition indicators at finer spatial detail
GRACE / GRACE-FO	Terrestrial water storage	Large-scale water-storage changes; not field-level groundwater readings
DATA ACCESS	NASA Earthdata / relevant mission portals / documented APIs; access and licensing must be checked per product.	

05 / STATISTICAL METHODS

Detect trends without overstating what they mean

MANN–KENDALL TEST

A non-parametric test for evidence of a monotonic trend in a time series. Account for seasonality and serial correlation where relevant.

SEN'S SLOPE

A robust estimate of the median rate of change, reported with its unit and time basis (for example, mm per year).

STANDARDIZED ANOMALY

Compare a value with a defined historical baseline. A Z-score expresses distance in standard-deviation units—not a direct event probability.

RECOMMENDED ANALYSIS SEQUENCE

Define region and time window → inspect coverage and quality flags → aggregate consistently → select baseline → calculate trend / anomaly → test robustness → report uncertainty and limitations.

06 / ILLUSTRATIVE CASE STUDY

Rainfall trend investigation: Sylhet

WORKFLOW

- **Define the study area**
  Use a documented Sylhet boundary or user-drawn polygon.
- **Build a comparable series**
  Aggregate the same months and spatial mask across years.
- **Test the trend**
  Report trend direction, Sen's slope, p-value and data coverage.

IMPORTANT DISTINCTION

Historical rainfall analysis can identify patterns and anomalies. It does not, by itself, forecast next June's rainfall or establish flood probability. Flood risk also depends on river levels, drainage, topography, exposure and forecasts.

All numerical values shown in an implementation must come from a documented run; no sample numbers are asserted here.

07 / ILLUSTRATIVE CASE STUDY

Salinity pressure and crop planning: Satkhira

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- **Satellite screening**
  Use appropriate Landsat surface-reflectance bands and quality masks to derive a candidate indicator.
- **Ground validation**
  Compare the indicator with field electrical-conductivity (EC) samples collected at suitable depths and dates.
- **Agronomic interpretation**
  Compare validated salinity levels with crop tolerance guidance, soil type, water availability and local extension advice.

A satellite salinity index is a proxy, not a direct EC measurement. Do not publish a yield-loss probability or crop prescription without a calibrated, locally validated model.

08 / EVIDENCE DOSSIER

Make every answer inspectable and reproducible

ORION EVIDENCE DOSSIER / TEMPLATE

Question	User's exact question
Area	Boundary name / geometry / area
Dataset	Product, collection, version, provider
Period	Observation window + baseline period
Processing	Masking, resampling, aggregation, units
Method	Formula / test / parameter choices
Result	Metric, units, direction, statistical context
Uncertainty	Coverage gaps, confidence interval if computed
Reproducibility	Run ID, code version, notebook / parameters

TRUST CHECKLIST

- **Source traceability**
  Can a reviewer locate the exact product and dates?
- **Method transparency**
  Are formula, units and parameters stated?
- **Honest uncertainty**
  Are limitations and data gaps visible?
- **Reproducibility**
  Can the same inputs regenerate the result?

09 / FROM TREND TO ACTION

Translate evidence into careful, useful next steps

Orion should support decisions—not replace local authorities, agronomists, hydrologists or emergency services.

DETECT

Identify a statistically supported signal: rainfall shift, vegetation stress, temperature anomaly or moisture deficit.

INTERPRET

Check exposure, local context, data quality and whether the indicator is relevant to the decision.

RESPOND

Offer validated options, monitoring priorities and links to trusted local guidance.

EXAMPLE: AGRICULTURAL DECISION SUPPORT

If validated field measurements indicate elevated soil salinity, the system may surface locally approved salt-tolerant varieties and management practices as options for expert review. It should state the evidence, location limits and conditions—not issue a universal prescription.

10 / GLOBAL RELEVANCE

One architecture, many Earth-system questions

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- **Bangladesh**
  Flood and rainfall variability, soil moisture, salinity screening and agricultural resilience.

- **Sahel region**
  Rainfall anomalies, vegetation stress and drought monitoring.

- **Europe**
  Heat extremes, land-surface temperature and vegetation condition.

H3 hexagonal indexing can provide a consistent spatial aggregation layer for visualization and comparison.

11 / DELIVERY ROADMAP

Build, validate, demonstrate, improve

A credible prototype is strongest when it shows a working path from question to a reproducible result.

PHASE 1
Data access + boundary selection
Confirm products, region, periods and data-quality rules.

PHASE 2
Scientific engine
Implement ingestion, processing, metrics and tests.

PHASE 3
Conversational interface
Map user questions to validated analysis templates.

PHASE 4
Validation + demo
Compare with references / field evidence; document limitations.



Explore Earth. Discover tomorrow.